

Technology Stack

Date	2 October 2026
Team ID	SWTID-2026-8368
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Jinja2 templates; Bangers and Poppins fonts	Server-rendered pages keep the app simple; responsive glass-style form with pill options
2	Backend / Server-Side	Python 3.14, FastAPI, Uvicorn	Fast to build, typed, automatic API docs at /docs; team knows Python
3	Database / Data Storage	No database. Files in static/panels (PNG) and static/exports (PDF)	A comic is generated and downloaded in one session; nothing needs to be queried later
4	Cloud / Hosting / Deployment	Runs locally with Uvicorn (Apple Silicon MPS, CUDA or CPU)	Avoids GPU hosting cost; Docker and a cloud GPU are listed in the future plan
5	Version Control & CI/CD	Git and GitHub; pytest (24 tests) run locally	Team collaboration through pull requests; CI is planned
6	Third-Party APIs / Other Tools	Google Gemini (google-generativeai); Stable Diffusion v1.5 via diffusers, PyTorch, transformers, accelerate; fpdf2; Pillow; python-dotenv	Gemini free tier for text; local model for images at no per-image cost; fpdf2 for PDF; dotenv for secrets